Portfolio

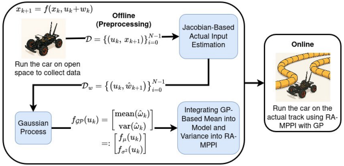
Hanif Edma

Research, engineering projects, and web apps

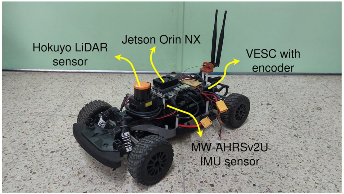
See it live: https://hanifedma.com/portfolio

Performance-Enhanced Risk-Aware MPPI using Gaussian Process

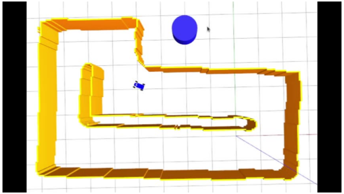
RESEARCH



Method framework



F1TENTH platform



Control demo

Performance-Enhanced Risk-Aware MPPI using Gaussian Process

RESEARCH

Published | 1st author | IEEE Access | SCI Q1 | 2025

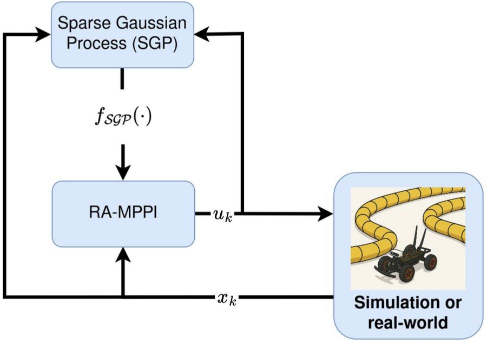
- Trains a Gaussian Process to learn the gap between a robot's nominal model and its true dynamics.
- Captures both the mean deviation and its uncertainty, feeding them into a risk-aware MPPI controller.
- Improves tracking accuracy and safety over baseline controllers.
- Validated in simulation and on real hardware (F1TENTH and Clearpath Jackal).

Images, video & explanation: https://gp-ramppi.github.io/

Paper (IEEE Access): https://doi.org/10.1109/ACCESS.2025.3640166

Robust RA-MPPI using Online Learning

RESEARCH



Control scheme

Real-world demo

Robust RA-MPPI using Online Learning

RESEARCH

Submitted to IJCAS (Springer) | 1st author

- Combines risk-aware MPPI control with online learning.
- A Sparse Gaussian Process identifies and quantifies model–reality discrepancies during operation.
- The controller uses this learned uncertainty to improve robustness and collision avoidance.
- Demonstrated on an autonomous vehicle with better trajectory tracking than standard MPPI.

Images, video & explanation: https://sgp-mppi.github.io/

ESP32 IoT Medical Gateway

ENGINEERING PROJECTS

ESP32 IoT Medical Gateway

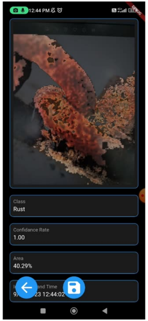
ENGINEERING PROJECTS

Internship | Team of 2 | Embedded / IoT | ESP32 | 2022

- Built as an IoT Engineer Intern at DoctorTool (PT Medifa Infoyasa Suryantara).
- Developed an ESP32-based IoT gateway bridging medical instruments to the cloud.
- Connects thermometers, oximeters, and biometric sensors.
- Supports over six device categories.
- Built as a team of 2 people.

Corrosion Detection App (Computer Vision)

ENGINEERING PROJECTS



In-app rust detection

1st place at SciHackfest 2023

Corrosion Detection App (Computer Vision)

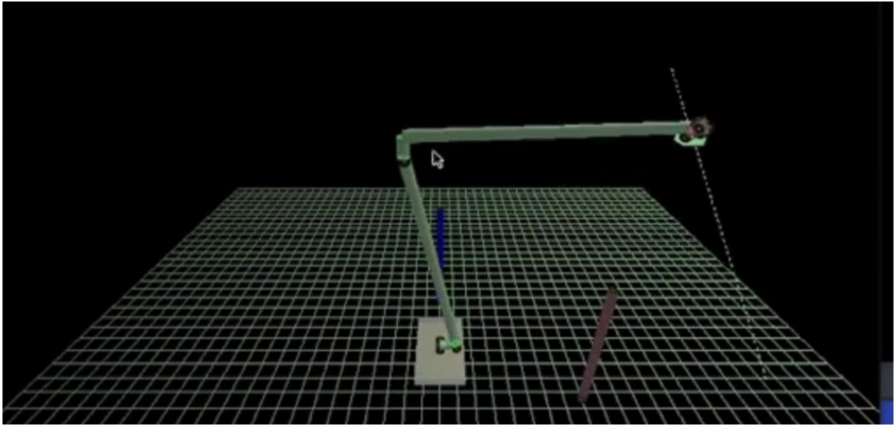
ENGINEERING PROJECTS

1st place · Hackathon · Prize 25M IDR (~$1,600) · Team of 3 · Flutter · Computer Vision · 2023

- Built a computer-vision system for corrosion detection.
- Shipped the Android app with Flutter (also runs on Windows and Linux).
- Delivered within a one-month timeframe.
- Won 1st place at SciHackfest 2023 (a hackathon held by PT Sucofindo) — prize 25 million IDR (~$1,600), as a team of 3.

6-DOF Robot Manipulator Simulation (OpenGL)

ENGINEERING PROJECTS



6-DOF Robot Manipulator Simulation (OpenGL)

ENGINEERING PROJECTS

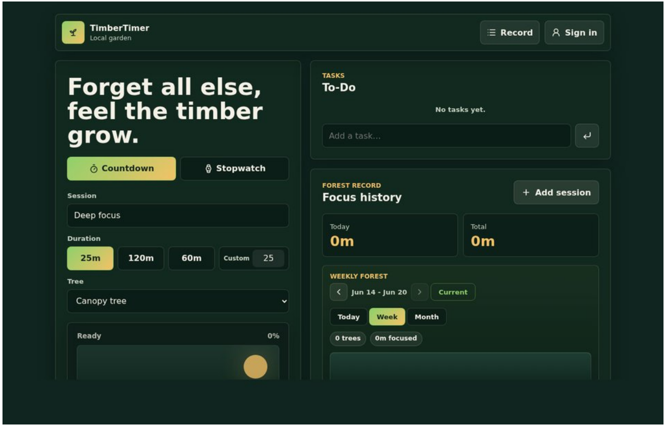
University project | Robot kinematics | C / OpenGL | Universitas Indonesia | 2022

- Real-time 3D visualization of a 6-DOF robot arm, built in C with OpenGL (GLUT).
- Implements forward kinematics and Jacobian pseudoinverse-based inverse kinematics for end-effector position control.
- Generates linear point-to-point trajectories between commanded poses.
- Streams joint commands to the physical robot over serial communication.

Source code: https://github.com/hanifedma/robot-6dof-opengl

TimberTimer

WEB APPS



TimberTimer

WEB APPS

Web app | Vanilla JS | Supabase | PWA

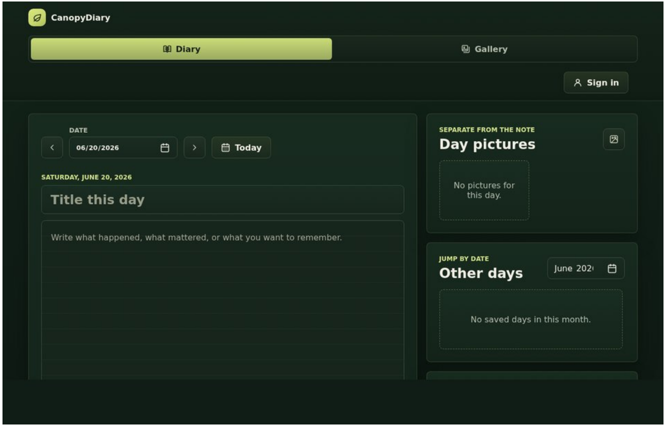
- Minimalist, jungle-themed focus timer that grows a virtual forest — one tree per completed session.
- Rest stopwatch, editable session history, and daily and cumulative stats.
- Local-first storage with optional cross-device sync (Supabase + Google sign-in).
- Installable as a Progressive Web App; built with vanilla JavaScript, HTML, and CSS.

Live app: https://hanifedma.com/timbertimer

Source code: https://github.com/hanifedma/timbertimer

CanopyDiary

WEB APPS



CanopyDiary

WEB APPS

Web app | Vanilla JS | Supabase | PWA

- Lightweight, local-first diary web app with dated notes.
- Date-sorted photo gallery kept separate from the written notes.
- Autosave and manual PDF export for selected date ranges.
- Optional cloud sync via Supabase + Google sign-in; built with vanilla JavaScript, HTML, and CSS.

Live app: https://hanifedma.com/canopydiary

Source code: https://github.com/hanifedma/canopydiary